Invoice Extraction

Pipeline Evaluation Report — 50 Documents (Interim)

HunyuanOCR + Qwen3-VL + LAYA system · MLX reader candidate · PaddleOCR candidates · degradation & super-resolution study

2026-10-03 · Corpus: synthetic val-50 · Hardware: Apple M5 Max (36 GB) · Status: NOT PRODUCTION READY · INTERIM: Paddle arms at reduced n

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Executive summary

Four evaluation arms (two readers × three input conditions) were tracked on the same 50 validation invoices across 10 vendors. The primary system — **Pipeline A on original inputs** — achieves 0.56 invoice exact-match and 0.84 critical-field exact-match with **zero hallucinated fields**, 0.99 citation coverage, and micro-F1 0.97. It fails production acceptance on whole-invoice correctness (target 0.90), critical-field correctness (0.84 vs 0.995), and latency (p95 30.6 s vs 10 s).

Finding. Super-resolution does not help: the OmniSR arm scores *below* the degraded arm on CER, coverage, and micro-F1. SR is rejected for this operating point.

Finding. The MLX 4B reader fails outright on 64% of documents (malformed JSON); where it runs, transcription matches Pipeline A but extraction recall collapses (0.30 vs 0.95).

Finding. Degraded input collapses every metric (field recall 0.21) — robustness to poor scans is the largest quality gap after latency.

Finding. Paddle interim (n=30): PP-StructureV3 reads the words (CER 0.309 vs HunyuanOCR 0.264) but loses the layout: reading-order score 0.17 vs 0.74, micro-F1 0.823 vs 0.972, invoice EM 0.10 vs 0.56. Header IDs and charge rows suffer most — table structure and block order are the gap.

1 Methodology

1.1 Dataset

50 invoices sampled from the 1,000-document validation split (layout family *layout_03*), stratified 5 per vendor across 10 vendors with natural field-presence variation. The validation split is sea-freight only and uses a held-out layout family, so results measure generalisation rather than template memorisation. Ground truth comprises per-field values, ordered text regions with bounding boxes, and paired degraded (3× downscale + JPEG) images for the robustness arms.

1.2 Pipelines under test

- **Pipeline A (primary).** Ingest: HunyuanOCR Q8_0 GGUF on Apple Metal (Docling layout/tables for digital PDFs). Reader: Qwen3-VL-8B-Instruct via Ollama, schema-constrained. Semantics: fine-tuned LAYA Choice vote + charge lexicon. Safety: evidence guard, hallucination blanking, citation boxes, REVIEW routing.
- **Pipeline B (candidate).** Same ingest and semantics as A; reader replaced by a Qwen3-4B MLX LoRA model fine-tuned on 3,000 corpus pairs (1,200 iterations, Mac Metal). All other stages identical.
- **Pipeline C (exploratory, unscored).** Docling (CPU) layout + tables + Tesseract OCR as ingest, Ollama reader otherwise. Evaluated for feasibility only — throughput too low for the 50-document track (0 docs).
- **Pipeline D (Paddle).** Ingest: PP-StructureV3 (PP-DocLayoutV3 + PP-OCRv6) with reading-ordered line output; reader, LAYA semantics, and safety stages identical to Pipeline A, so D-vs-A isolates ingest quality. Runs on Mac CPU under paddlepaddle 3.3.1.
- **Pipeline E (Paddle VL).** Same as D with PaddleOCR-VL-1.6 block parsing as the primary transcription leg (StructureV3 fallback). Slowest arm (~2 min/doc on CPU).

1.3 Input conditions

Code	Condition	Description
orig	Original	Rendered invoice at native resolution (1275×1650).
lr	Degraded (low-resolution)	Ground-truth paired variant: 3× downscale + JPEG compression.
sr	Super-resolved	Degraded variant restored with the fine-tuned OmniSR ×2 model on MPS.

1.4 Metric glossary

Metric	Definition
CER / WER	Character / word error rate between ground-truth and system transcription; reported raw and with approved normalisation (Unicode, whitespace, OCR markup). Lower better.
Coverage	Share of ground-truth tokens recovered. Hallucination (token): share of generated tokens with no ground-truth support. Lower hallucination better.

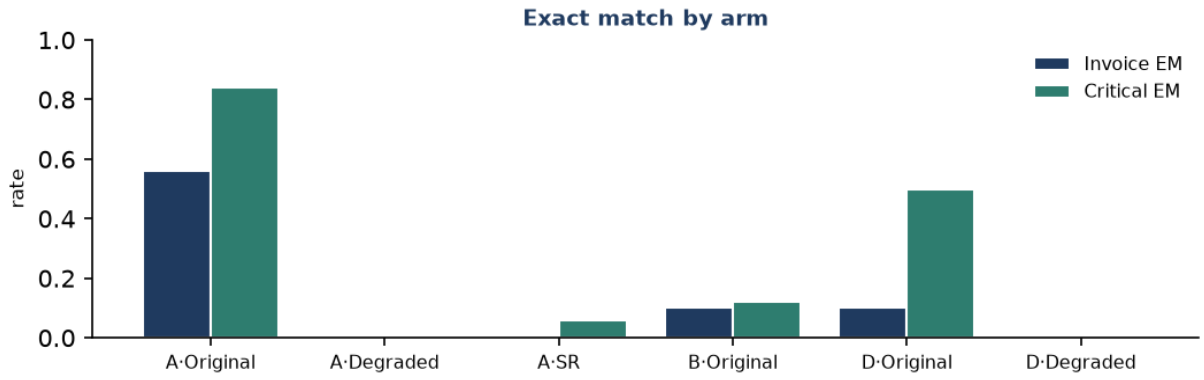
Metric	Definition
Field P / R / F1 / F2	Micro-averaged precision, recall, F1, and recall-weighted F2 over the 31 canonical fields. F2 is retained alongside F1 per the evaluation spec.
Field accuracy	Support-weighted exact-match rate over all field slots.
Invoice EM	Share of invoices with *every* field correct. Critical EM: share with all eight critical fields correct (Invoice No/Date, Vendor, Container, MAWB, HAWB, HBL, Weight).
Field hallucination	Share of non-empty predictions with no document evidence (evidence-guard verdict). Must be ~0 for production.
NED	Normalised edit distance of transcription vs ground truth. Reading order: pairwise ordering accuracy of ground-truth regions in the transcription.
Latency / citations	End-to-end seconds (mean, p95) with ingest/LLM/cite stages; citation coverage = predicted values with a measured bounding box.

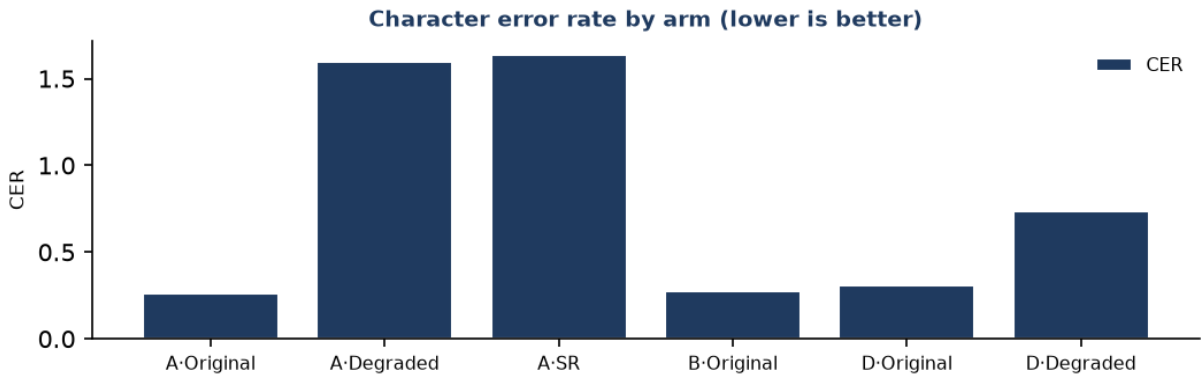
2 Results

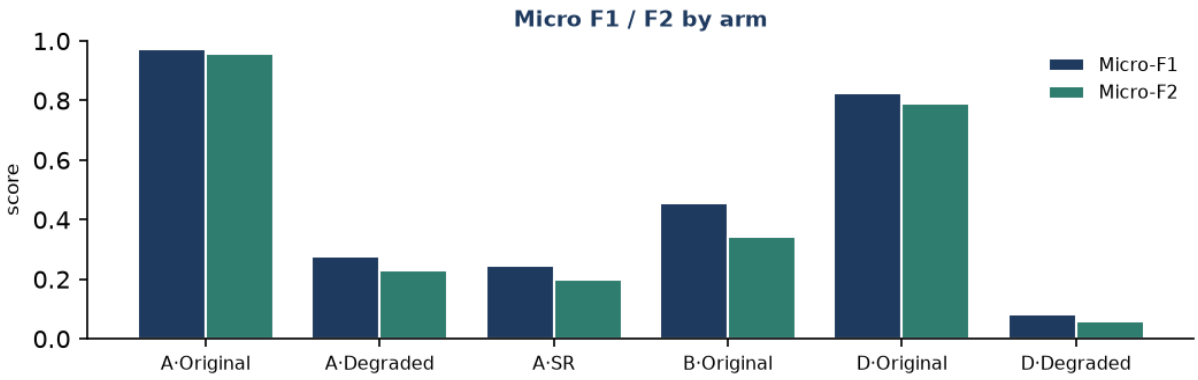
2.1 Arm comparison (n per arm)

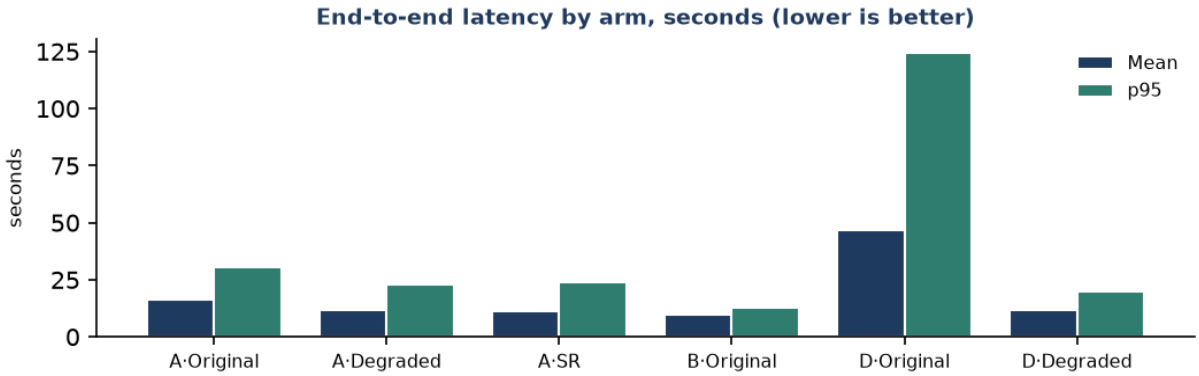
Evaluation arm	n	CER ↓	WER ↓	F1	F2	Acc	Inv EM	Crit EM	Cover	t-Hall↓	f-Hall↓	NED ↓	p95	Cite
Pipeline A · Original	50	0.264	0.245	0.972	0.956	0.975	0.56	0.84	0.978	0.198	0.000	0.208	30.6 s	0.99
Pipeline A · Degraded (LR)	50	1.598	1.004	0.277	0.230	0.623	0.00	0.00	0.370	0.690	0.000	0.517	23.1 s	0.06
Pipeline A · Super-Resolved (SR)	50	1.635	2.131	0.246	0.198	0.608	0.00	0.06	0.280	0.764	0.000	0.588	23.8 s	0.14
Pipeline B · Original	50	0.273	0.260	0.456	0.344	0.679	0.10	0.12	0.966	0.208	0.000	0.214	12.9 s	0.35
Pipeline D · Original	30	0.309	0.344	0.823	0.791	0.896	0.10	0.50	0.910	0.219	0.000	0.248	124.3s	0.98
Pipeline D · Degraded (LR)	30	0.738	0.947	0.081	0.059	0.557	0.00	0.00	0.070	0.918	0.000	0.735	20.1 s	0.02

↓ lower is better. t-Hall = token hallucination; f-Hall = field hallucination; Cite = citation coverage; p95 = end-to-end seconds.









2.2 Reading the results

Pipeline A on originals dominates every quality metric while keeping field hallucination at zero — the safety property the system is designed around. The cost is latency: HunyuanOCR reloads weights per call and the 8B reader averages ~6 s per invoice. Pipeline B shares A's transcription (CER 0.27 vs 0.26) but its 4B reader fails on 64% of documents, which also explains its low citation coverage (0.35). The degraded arms show that neither the OCR nor the SR stage survives 3× downscaling: coverage falls below 0.40 and extraction recall below 0.21.

Pipeline D is the ingest-controlled comparison: identical reader, LAYA, and safety stages, with PP-StructureV3 replacing HunyuanOCR. At n=30 the OCR itself is close (CER 0.309 vs 0.264, coverage 0.910 vs 0.978), but flattening the page to unordered lines costs the reader dearly: reading-order 0.17 vs 0.74, micro-F1 0.823 vs 0.972, invoice EM 0.10 vs 0.56, critical EM 0.50 vs 0.84. Per-field gaps concentrate on Bill of Lading, Container Number, and charge rows — values detached from labels. On degraded inputs D collapses harder than A (coverage 0.070 vs 0.370): PP-OCR detection fails on 3× downscaled scans where HunyuanOCR still recovers a third of tokens.

3 Per-field detail — Pipeline A, original inputs

Canonical field	EM	Precision	Recall	F1	F2
Invoice No	1.00	1.00	1.00	1.00	1.00
Invoice Date	1.00	1.00	1.00	1.00	1.00
Container Number	0.94	1.00	0.94	0.97	0.95
MAWB No	1.00	0.00	0.00	0.00	0.00
HAWB No	1.00	0.00	0.00	0.00	0.00
Vendor Name	1.00	1.00	1.00	1.00	1.00
Bill of Lading / HBL	0.86	1.00	0.86	0.92	0.88
WEIGHT	1.00	1.00	1.00	1.00	1.00
Sea Freight	0.92	1.00	0.76	0.87	0.80
Air Freight	0.98	1.00	0.93	0.96	0.94
Terminal Handling Charges	1.00	1.00	1.00	1.00	1.00
Airway WayBill Charges	1.00	1.00	1.00	1.00	1.00
BL Charges	0.96	1.00	0.91	0.95	0.93

Canonical field	EM	Precision	Recall	F1	F2
ENS Charges	1.00	1.00	1.00	1.00	1.00
Seal Charges	1.00	1.00	1.00	1.00	1.00
VGM	1.00	1.00	1.00	1.00	1.00
EDI Fees	1.00	1.00	1.00	1.00	1.00
DAP Charges	0.88	1.00	0.67	0.80	0.71
Other Charges	0.94	0.95	0.90	0.93	0.91
AMS CHARGES	1.00	1.00	1.00	1.00	1.00
Handlign Charges	1.00	1.00	1.00	1.00	1.00
Pre Carriage Haulage	1.00	1.00	1.00	1.00	1.00
Ex Rate	1.00	1.00	1.00	1.00	1.00
Misc	0.96	1.00	0.83	0.91	0.86
Transportation	0.96	1.00	0.85	0.92	0.87
Warehouse charges Port	1.00	1.00	1.00	1.00	1.00
Examination charges	0.96	1.00	0.83	0.91	0.86
Fuel Charges	0.94	1.00	0.88	0.94	0.90
Cargo Screening	0.94	1.00	0.81	0.90	0.84
Agency Charges	1.00	1.00	1.00	1.00	1.00
Shipping bill charges	0.98	1.00	0.95	0.97	0.96

4 Acceptance statement

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Metric**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=7, greek=0, italic=0, link=[], rise=0, text='Metric', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Production target**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=7, greek=0, italic=0, link=[], rise=0, text='Production target', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Actual (A/orig)**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=7, greek=0, italic=0, link=[], rise=0, text='Actual (A/orig)', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Status**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=7, greek=0, italic=0, link=[], rise=0, text='Status', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Invoice EM	≥ 0.90	0.56	FAIL
Critical-field EM	≥ 0.995	0.84	FAIL
Field hallucination	≤ 0.005	0.000	PASS
JSON schema compliance	1.00	1.00	PASS
p95 latency	≤ 10 s	30.6 s	FAIL
Throughput	≥ 1 page/s	0.06 page/s	FAIL
Citation coverage	measured	0.99	MEASURED

Finding. Verdict: NOT PRODUCTION READY. Blockers in priority order: critical-field correctness, whole-invoice correctness, latency/throughput, zero straight-through processing (all REVIEW), no production database, synthetic-only evidence.

5 Limitations & reproduction

- Validation split is sea-freight only with a single layout family; air-freight and layout-04 generalisation are unmeasured here (frozen test numbers exist separately).
- Table-structure F1 is NOT MEASURED (corpus has no table ground truth); chart metrics are NOT APPLICABLE (no charts). Pipeline C unscored for throughput reasons.
- Interim arms: D at n=30 (top-up 20 pending); E excluded (partial, rerunning after an OOM kill). D/orig latency is overstated — part of the track ran under CPU contention with the VL arm (ingest p95 115 s vs ~20 s uncontended).
- All training is currently paused (Qwen3-VL-8B QLoRA before iteration 1; MLX 4B reader at iteration 1,200 with 64% failure).

- Reproduce: `scripts/track_pipelines.py --count 50` + `scripts/track_paddle.py` (paddle-env) → `scripts/aggregate_track.py` → `scripts/build_pdf_report.py`. Raw rows: `work/track*/pred_*.jsonl`.